

# SSH Detection and Response in Microsoft Sentinel

Meridian Trust Bank training scenario  
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Report reference: MTB-LAB-2026-01 | Audience: SOC lead and portfolio reviewers

## Summary

Analysis of 291 simulated Linux log records identified a password-guessing sequence followed by successful authentication to opsadmin, an unsuccessful multi-account login pattern consistent with password spraying, and repeated backup service failures. The compromise scenario warrants priority escalation and investigation of privileged activity. No containment or recovery actions were performed as part of the documented exercise.

| Case                    | Assessment                                                            | Priority and response                                      |
|-------------------------|-----------------------------------------------------------------------|------------------------------------------------------------|
| LAB-01<br>203.0.113.77  | 14 failures, then 1 success<br>True positive in the scenario          | High<br>Escalate for host and account containment          |
| LAB-02<br>10.20.14.9    | 30 failures for svc-backup<br>Benign explanation supplied by exercise | Low after owner verification<br>Repair service credentials |
| LAB-03<br>198.51.100.23 | 11 failures across 11 usernames<br>Suspected password spray           | Medium<br>Investigate scope and monitor targeted accounts  |

## Scope and current status

The environment is a fictional bank lab, not a production bank or a customer engagement. One host, mtb-app01, is represented. I configured Sentinel and used the MeridianLogs inline datatable to investigate the sample data. A captured MeridianLogs\_CL count returned zero for the selected range. Successful persistent ingestion, scheduled alert generation, incident closure, and a workbook are not yet demonstrated.

Event times below retain the log clock: 20 July 06:17:00 through 22 July 05:17:00, a 47-hour span described in the guide as a 48-hour pack. The original lines omit the year and timezone; the September report date is the analysis date, not the attack date.

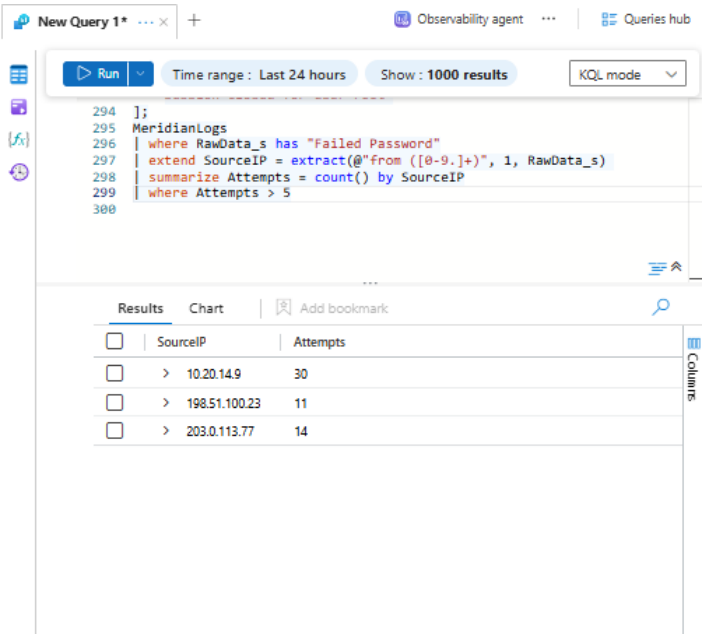
## Detection

I filtered SSH authentication failures, extracted source IP addresses and counted attempts. The sample contains 68 failed and 50 accepted-password records; the remaining 173 records include session and scheduled-job activity. Three sources exceeded the greater-than-five threshold, accounting for 55 of the 68 failures. A threshold match identifies a case for investigation, not proof of compromise.

### Implemented exploratory query

The query runs after the supplied `let MeridianLogs = datatable(...)` definition. This readable equivalent uses contains for the message phrase. Figure E1 shows the original executed threshold query and results.

```
MeridianLogs
| where RawData_s contains "Failed password"
| extend SourceIP = extract(@"from ([0-9.]+)", 1, RawData_s)
| where isnotempty(SourceIP)
| summarize Attempts = count() by SourceIP
| where Attempts > 5
| order by Attempts desc
```



E1 Inline threshold query returns three source IPs. These are query results, not Sentinel incident records.

### Scheduled detection specification

The guide proposes Brute Force - Multiple Failed SSH Logins with Medium severity, a five-minute schedule, a one-hour lookback and more than five failures in ten-minute bins. That scheduled rule is a next step, not an implemented control in this report. The inline query above totals the whole dataset and has no time window. Alert and incident creation must be verified separately.

# Incident triage

## LAB 01 Password guessing followed by successful login

Source 203.0.113.77 generated 14 failures against opsadmin on mtb-app01 from 22 July 02:13:00 to 02:14:24. An accepted-password event followed at 02:14:35, 95 seconds after the first failure. E2 verifies the failure-to-success sequence. The exercise identifies the activity as a compromise; the log pattern supports a high-priority investigation, but does not reveal how the successful password was obtained.

| Log time on 22 July  | Evidence and interpretation                                                |
|----------------------|----------------------------------------------------------------------------|
| 02:13:00 to 02:14:24 | 14 failures; original log lines 264–277.                                   |
| 02:14:35             | Password accepted and session opened for opsadmin; lines 278–279.          |
| 02:16:35             | sudo records opsadmin invoking /usr/bin/cat /etc/shadow as root; line 280. |

Supplementary raw-log review links the privileged command by host, account and timing. E2 alone does not show it because that line has no source IP. The command indicates possible credential-hash exposure; command output, successful file extraction and outbound exfiltration are not established. Lateral movement and broader impact remain unknown.

New Query 1\*

+

Observability agent

Save

Share

...

Quer

Run

Time range: Last 24 hours

Show: 1000 results

KQL mode

```
293 "Jul 22 05:17:00 mtb-app01 CROW[23818]: pam_unix(cron:session): session closed for u
root"
294 };
295 MeridianLogs
296 | where RawData_s contains "203.0.113.77"
297 | extend Action = case(
298   RawData_s contains "Accepted password", "SUCCESS",
299   RawData_s contains "Failed password", "failed",
300   "other"
301 )
302 | project RawData_s, Action
```

Results

Chart

Add bookmark

| RawData_s                                                                                                         | Action  |
|-------------------------------------------------------------------------------------------------------------------|---------|
| > Jul 22 02:13:00 mtb-app01 sshd[15320]: Failed password for opsadmi...                                           | failed  |
| > Jul 22 02:13:06 mtb-app01 sshd[15320]: Failed password for opsadmi...                                           | failed  |
| > Jul 22 02:13:13 mtb-app01 sshd[15320]: Failed password for opsadmi...                                           | failed  |
| > Jul 22 02:13:19 mtb-app01 sshd[15320]: Failed password for opsadmi...                                           | failed  |
| > Jul 22 02:13:26 mtb-app01 sshd[15320]: Failed password for opsadmi...                                           | failed  |
| > Jul 22 02:13:32 mtb-app01 sshd[15320]: Failed password for opsadmi...                                           | failed  |
| > Jul 22 02:13:39 mtb-app01 sshd[15320]: Failed password for opsadmi...                                           | failed  |
| > Jul 22 02:13:45 mtb-app01 sshd[15320]: Failed password for opsadmi...                                           | failed  |
| > Jul 22 02:13:52 mtb-app01 sshd[15320]: Failed password for opsadmi...                                           | failed  |
| > Jul 22 02:13:58 mtb-app01 sshd[15320]: Failed password for opsadmi...                                           | failed  |
| > Jul 22 02:14:05 mtb-app01 sshd[15320]: Failed password for opsadmi...                                           | failed  |
| > Jul 22 02:14:11 mtb-app01 sshd[15320]: Failed password for opsadmi...                                           | failed  |
| > Jul 22 02:14:18 mtb-app01 sshd[15320]: Failed password for opsadmi...                                           | failed  |
| > Jul 22 02:14:24 mtb-app01 sshd[15320]: Failed password for opsadmi...                                           | failed  |
| > Jul 22 02:14:35 mtb-app01 sshd[15320]: Accepted password for ops...                                             | SUCCESS |
| RawData_s Jul 22 02:14:35 mtb-app01 sshd[15320]: Accepted password for opsadmin from 203.0.113.77 port 49215 ssh2 |         |
| Action SUCCESS                                                                                                    |         |

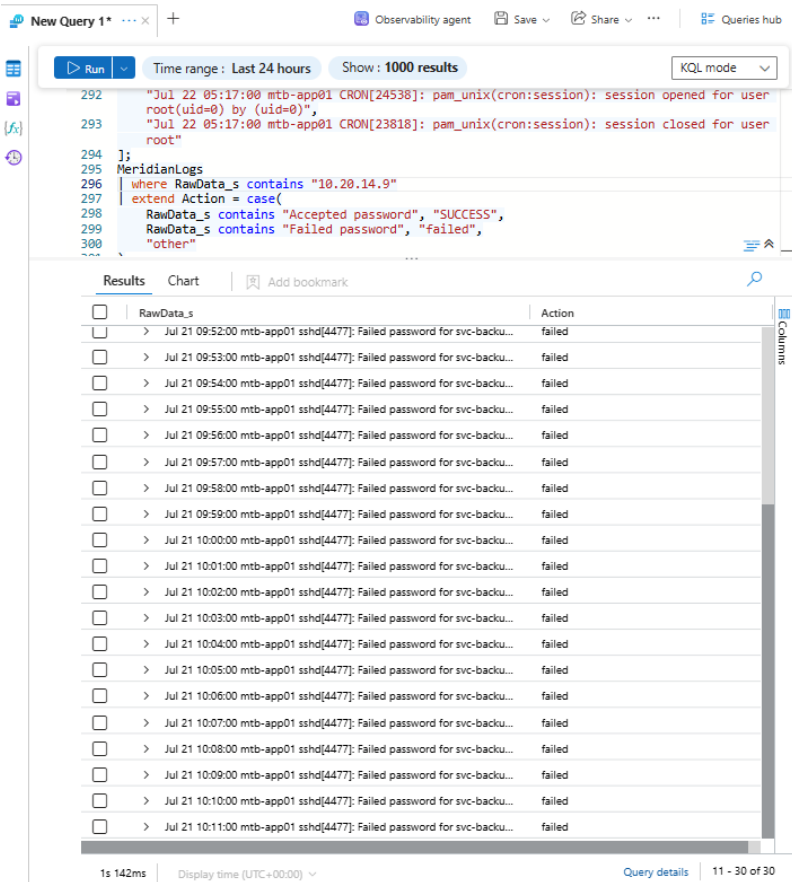
E2 Successful authentication after repeated failures. The expanded result identifies opsadmin and the source IP.

Disposition: True positive within the lab scenario. Proposed priority: High. Recommend escalation, preservation of evidence and authorised host/account containment. No production incident was closed or remediated.

LAB 02 Repeated backup service authentication failures

Source 10.20.14.9 generated 30 failed logins for svc-backup on mtb-app01 between 21 July 09:42:00 and 10:11:00. The records occur exactly one minute apart, with the same source port and process identifier. No accepted-password event from this IP appears in the supplied sample. E3 shows the repetitive failure pattern; the raw log establishes the complete interval.

This pattern is consistent with an automated job using an outdated credential. The guide supplies the stale-backup-password explanation; there is no service-owner confirmation or configuration evidence in the exercise. An internal address and failed authentication alone do not establish that activity is safe.



E3 Thirty rows are returned for the backup source. The visible rows show regular failures; the full interval is confirmed in log lines 144–173.

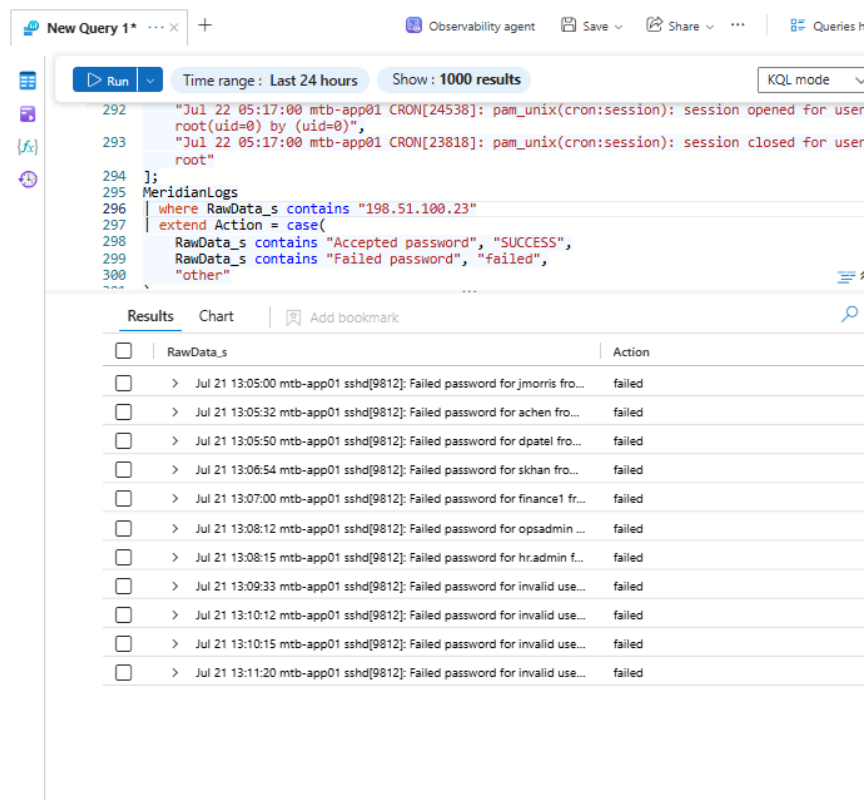
Disposition: Benign for the training scenario, based on the guide’s context. In production, retain an undetermined assessment until the service owner verifies the job, host and credential change. If expected activity is confirmed and the rule detected the pattern correctly, Benign Positive is appropriate; use False Positive when the alert logic or data is incorrect.

Recommended action: correct the job credential, verify a successful backup run, and narrowly tune the confirmed service pattern if needed. Do not block the internal source or broadly exclude internal IPs simply because it has the highest failure count.

## LAB 03 Multi account login attempts

Source 198.51.100.23 generated 11 failed logins across 11 distinct usernames on mtb-app01 between 21 July 13:05:00 and 13:11:20. Seven entries target named accounts and four identify invalid usernames. No successful authentication from this source is present in the supplied sample. E4 shows all 11 failures.

One attempt per username from a common source is consistent with password spraying. The guide identifies this as a spray, but SSH logs do not record the attempted passwords, so reuse of one password cannot be proven from these records. The precise finding is a suspected password spray or other multi-account guessing attempt.



The screenshot displays a security tool interface with a KQL query and its results. The query is as follows:

```
292 "Jul 22 05:17:00 mtb-app01 CRON[24538]: pam_unix(cron:session): session opened for user  
293 root(uid=0) by (uid=0)",  
294 "Jul 22 05:17:00 mtb-app01 CRON[23818]: pam_unix(cron:session): session closed for user  
295 root"  
296  
297 MeridianLogs  
298 where RawData_s contains "198.51.100.23"  
299 extend Action = case(  
300 RawData_s contains "Accepted password", "SUCCESS",  
301 RawData_s contains "Failed password", "failed",  
302 "other"
```

The results table shows 11 failed login attempts for various usernames on mtb-app01:

| RawData_s                                                                  | Action |
|----------------------------------------------------------------------------|--------|
| > Jul 21 13:05:00 mtb-app01 sshd[9812]: Failed password for jmorris fro... | failed |
| > Jul 21 13:05:32 mtb-app01 sshd[9812]: Failed password for achen fro...   | failed |
| > Jul 21 13:05:50 mtb-app01 sshd[9812]: Failed password for dpatel fro...  | failed |
| > Jul 21 13:06:54 mtb-app01 sshd[9812]: Failed password for skhan fro...   | failed |
| > Jul 21 13:07:00 mtb-app01 sshd[9812]: Failed password for finance1 fr... | failed |
| > Jul 21 13:08:12 mtb-app01 sshd[9812]: Failed password for opsadmin ...   | failed |
| > Jul 21 13:08:15 mtb-app01 sshd[9812]: Failed password for hr.admin f...  | failed |
| > Jul 21 13:09:33 mtb-app01 sshd[9812]: Failed password for invalid use... | failed |
| > Jul 21 13:10:12 mtb-app01 sshd[9812]: Failed password for invalid use... | failed |
| > Jul 21 13:10:15 mtb-app01 sshd[9812]: Failed password for invalid use... | failed |
| > Jul 21 13:11:20 mtb-app01 sshd[9812]: Failed password for invalid use... | failed |

**E4 Eleven failed logins across different usernames from one source. No accepted-password event is returned for this source.**

Disposition: True positive attempted attack within the exercise. Proposed priority: Medium. No successful authentication was observed in this dataset; that does not establish that the source never succeeded elsewhere or outside the collection period.

Recommended action: search other hosts and identity logs for the source and targeted accounts, review later successes, and apply a scoped block or rate limit if operationally justified. Continue monitoring account-focused patterns to detect distributed attempts.

# MITRE ATTACK mapping

| MITRE   ATT&CK®                                                                                               |                                                                                                                  |                                                                                                               |                                                                                                           |                                                                                                                     |                                                                                                                     |                                                                                                                        |                                                                                                             |                                                                                                                        |                                                                                                                   |                                                                                                     |                                                                                                                    |                                                                                                           |                                                                                                              |                                                                                                                   |                                                                                                              | Matrices • Tactics • Techniques • Defenses • CTI • Resources • Benefactors • Contribute • Blog  Search  |  |
|---------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| Reconnaissance                                                                                                |                                                                                                                  | Resource Development                                                                                          | Initial Access                                                                                            | Execution                                                                                                           | Persistence                                                                                                         | Privilege Escalation                                                                                                   | Stealth                                                                                                     | Defense Impairment                                                                                                     | Credential Access                                                                                                 | Discovery                                                                                           | Lateral Movement                                                                                                   | Collection                                                                                                | Command and Control                                                                                          | Exfiltration                                                                                                      | Impact                                                                                                       |                                                                                                                                                                                                                                                                               |  |
| 12 techniques                                                                                                 |                                                                                                                  | 9 techniques                                                                                                  | 11 techniques                                                                                             | 20 techniques                                                                                                       | 22 techniques                                                                                                       | 13 techniques                                                                                                          | 36 techniques                                                                                               | 18 techniques                                                                                                          | 17 techniques                                                                                                     | 34 techniques                                                                                       | 9 techniques                                                                                                       | 17 techniques                                                                                             | 18 techniques                                                                                                | 9 techniques                                                                                                      | 15 techniques                                                                                                |                                                                                                                                                                                                                                                                               |  |
| Active Sensing               | Gather Victim Host Information  | Acquire Infrastructure       | Content Injection                                                                                         | BITS Jobs                                                                                                           | Account Manipulation               | Abuse Elevation Control Mechanism     | Access Token Manipulation  | Disable or Modify System Firewall     | Adversary-in-the-Middle          | Account Discovery  | Exploitation of Remote Services                                                                                    | Adversary-in-the-Middle  | Application Layer Protocol  | Automated Purification           | Account Access Removal                                                                                       |                                                                                                                                                                                                                                                                               |  |
|                                                                                                               |                                                                                                                  |                                                                                                               | Drive-by Compromise                                                                                       | Cloud Administration Command                                                                                        | Build or Logon Assistant                                                                                            | Access Token Manipulation             | Build Image on Host                                                                                         | Domain or Tenant Policy Modification  | Credentials from Password Store  | Blame Force        | Internal Spearphishing                                                                                             | Internal Spearphishing                                                                                    | Communication Through Removable Media                                                                        | Data Transfer Size Limits                                                                                         | Data Destruction            |                                                                                                                                                                                                                                                                               |  |
|                                                                                                               |                                                                                                                  |                                                                                                               | Exploit Public-Facing Application                                                                         | Command and Scripting Interpreter  | Boot or Logon Subsystem Execution  | Account Manipulation                  | Debugger Evasion                                                                                            | Downgrade Attack                                                                                                       | Exploitation for Credential Access                                                                                | Cloud Infrastructure Discovery                                                                      | Remote Service Session Hijacking  | Automated Collection                                                                                      | Data Encoding               | Defacement                       | Data Manipulation           |                                                                                                                                                                                                                                                                               |  |
|                                                                                                               |                                                                                                                  |                                                                                                               | External Remote Services                                                                                  | Container Administration Command                                                                                    | Boot or Logon Subsystem Execution  | Cloud Application Integration                                                                                          | Deauthenticate/Decode Files or Information                                                                  | Exploitation for Defense Impairment                                                                                    | Forward Authentication                                                                                            | Cloud Service Dashboard                                                                             | Remote Services                   | Clipboard Data                                                                                            | Data Obfuscation            | Defacement                       | Data Manipulation           |                                                                                                                                                                                                                                                                               |  |
|                                                                                                               |                                                                                                                  |                                                                                                               | Hardware Additions                                                                                        | Deploy Container                                                                                                    | Cloud Application Integration                                                                                       | Compromise Host Software Binary                                                                                        | Direct Volume Access                                                                                        | Exploitation for Stealth                                                                                               | Forge Web Credentials            | Cloud Storage Object Discovery                                                                      | Remote Services                   | Clipboard Data                                                                                            | Data Obfuscation            | Defacement                       | Data Manipulation           |                                                                                                                                                                                                                                                                               |  |
| Phishing for Information     | Gather Victim Org Information   | Develop Capabilities         | Phishing                 | EDM Administration Command                                                                                          | Create Account                     | Domain or Tenant Policy Modification  | Escape to Host                                                                                              | Event Triggered Execution             | Hide Artifacts                   | Multi-Factor Authentication Interception                                                            | Domain Trust Discovery                                                                                             | File and Directory Discovery                                                                              | Data from Cloud Storage                                                                                      | Estimation Over Physical Medium  | Endpoint Denial of Service  |                                                                                                                                                                                                                                                                               |  |
|                                                                                                               |                                                                                                                  |                                                                                                               | Replication Through Removable Media                                                                       | Exploitation for Client Execution                                                                                   | Create or Modify System Process    | Domain or Tenant Policy Modification  | Event Triggered Execution  | Indicator Removal                     | Modify Cloud Resource Hierarchy                                                                                   | Multi-Factor Authentication Interception                                                            | Domain Trust Discovery                                                                                             | File and Directory Discovery                                                                              | Data from Cloud Storage                                                                                      | Estimation Over Physical Medium  | Endpoint Denial of Service  |                                                                                                                                                                                                                                                                               |  |
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| Search Open Technical Data   | Search Open Technical Data      | Search Open Technical Data   | Valid Accounts           | Input Injection                                                                                                     | Event Triggered Execution          | Domain or Tenant Policy Modification  | Event Triggered Execution  | Indicator Removal                     | Modify Cloud Resource Hierarchy                                                                                   | Multi-Factor Authentication Interception                                                            | Domain Trust Discovery                                                                                             | File and Directory Discovery                                                                              | Data from Cloud Storage                                                                                      | Estimation Over Physical Medium  | Endpoint Denial of Service  |                                                                                                                                                                                                                                                                               |  |
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| Search Threat Vendor Data    | Search Threat Vendor Data       | Search Threat Vendor Data    | Valid Accounts           | Input Injection                                                                                                     | Event Triggered Execution          | Domain or Tenant Policy Modification  | Event Triggered Execution  | Indicator Removal                     | Modify Cloud Resource Hierarchy                                                                                   | Multi-Factor Authentication Interception                                                            | Domain Trust Discovery                                                                                             | File and Directory Discovery                                                                              | Data from Cloud Storage                                                                                      | Estimation Over Physical Medium  | Endpoint Denial of Service  |                                                                                                                                                                                                                                                                               |  |
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| Search Victim-Owned Website  | Search Victim-Owned Website     | Search Victim-Owned Website  | Valid Accounts           | Input Injection                                                                                                     | Event Triggered Execution          | Domain or Tenant Policy Modification  | Event Triggered Execution  | Indicator Removal                     | Modify Cloud Resource Hierarchy                                                                                   | Multi-Factor Authentication Interception                                                            | Domain Trust Discovery                                                                                             | File and Directory Discovery                                                                              | Data from Cloud Storage                                                                                      | Estimation Over Physical Medium  | Endpoint Denial of Service  |                                                                                                                                                                                                                                                                               |  |
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| Technique                          | Case and supporting behaviour                                      | Qualification                                                                 |
|------------------------------------|--------------------------------------------------------------------|-------------------------------------------------------------------------------|
| T1110.001<br>Password Guessing     | LAB-01: rapid repeated failures against opsadmin.                  | Passwords attempted are not recorded.                                         |
| T1078<br>Valid Accounts            | LAB-01: accepted authentication and a session for opsadmin.        | Malicious use is established by the scenario context and supporting sequence. |
| T1003.008<br>OS Credential Dumping | LAB-01: root-context command targeting /etc/shadow in the raw log. | Possible hash access, no proof of exfiltration or cracking.                   |
| T1110.003<br>Password Spraying     | LAB-03: a single source tries 11 usernames.                        | Consistent with spraying, identical password reuse is unverified.             |

## Remediation

The following actions are recommendations for an authorised response team. They were not executed against a live endpoint in this exercise.

### Priority response to LAB 01

The SOC lead should escalate the successful-login sequence and preserve authentication, sudo, session and available endpoint records. The endpoint team should assess the isolation of mtb-app01 and terminate the suspicious session while preserving evidence. The identity owner should restrict opsadmin and rotate its credential; the /etc/shadow command warrants assessment of other local credentials that may be exposed.

Investigators should review sudo privileges, authorised SSH keys, scheduled jobs, services and subsequent connections. Correlate other hosts before declaring the scope contained. Restore service only after persistence

checks, credential remediation and monitoring confirm recovery; a closed alert alone is not evidence of eradication.

### **Service repair and prevention**

For LAB-02, the backup owner should validate the job and repair its authentication without interrupting unrelated services. For LAB-03, the SOC should review targeted accounts and apply proportionate network controls. Longer-term measures include restricting SSH reachability, using managed keys and MFA where supported, reducing unnecessary sudo access and adding detections for privileged file access and successful logins after failures.

## Detection limitations

### Counting and event time

The demonstrated fallback query counts failures over the entire static sample. It cannot support a claim of detecting more than five attempts within ten minutes. The importer leaves TimeGenerated at collection/arrival time and places the original timestamp in EventTime. Bulk import therefore compresses older activity into recent ingestion bins. Original event times must be parsed and used to evaluate attack timing; ingestion time remains useful for checking delivery.

The raw records omit year and timezone. The importer assumes the current year and appends Z; that is a lab conversion assumption, not proof that the source clock was UTC. Timeline and workbook labels must preserve this distinction.

### Coverage and tuning

Fixed ten-minute bins can split a burst across a boundary. A source-IP threshold can miss distributed attempts, slow guessing and activity spread across hosts. Shared addresses and stale service credentials can produce noise. A companion rule should assess distinct usernames, account totals over longer periods, and failures followed by success. Tuning requires benign test cases as well as known attack cases.

IP-only pivots omit session and sudo records that lack the IP. Correlate host, username and event time, and verify session identifiers where available. A query labelled SUCCESS means authentication was accepted; it does not independently prove attacker ownership, privilege level or business impact.